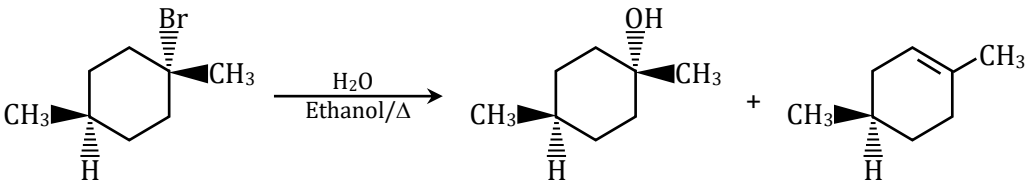
RACE
Regular Analysis Through Continuous Exercise

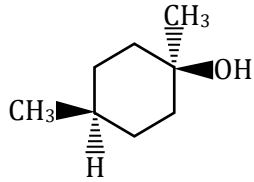
Alkyl Halides or Haloalkanes : RACE-01

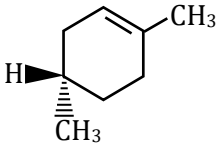
Substitution Rxⁿ, Elimination Rxⁿ, SN vs E, Nucleophilicity vs Basicity

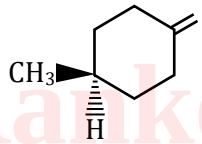
1. Consider the following E1/S_N1 reaction

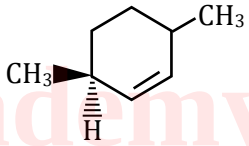


The missing product(s)

(1) 

(2) 

(3) 

(4) 

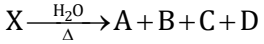
(A) 1, 2 and 3 (B) 3 and 4 (C) 2 and 3 (D) 1, 2, 3 and 4

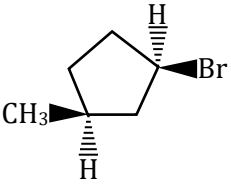
CHL158

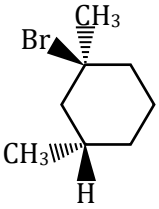
2. Select the reagent that will yield the greater amount of substitution on reaction with $CH_3–CH_2–Br$:
(A) CH_3CH_2OK in dimethyl sulfoxide (DMSO)
(B) $(CH_3)_3$ COK in dimethyl sulfoxide (DMSO)
(C) Both (A) and (B) will give comparable amounts of substitution
(D) Neither (A) nor (B) will give any amount of substitution

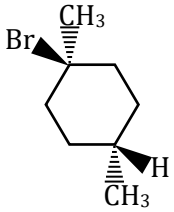
CHL159

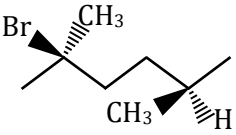
3. Under the specified conditions, substrate X undergoes substitution and elimination reactions to give products A –D. A and B are stereoisomers, but not enantiomers. C and D are enantiomers. A is not an isomer of C. Which of the following could be the starting material X?



(I) 

(II) 

(III) 

(IV) 

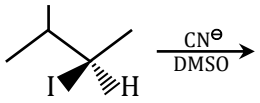
(A) (I) (B) (II) (C) (III) (D) (IV)

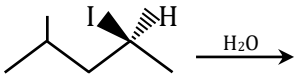
CHL160

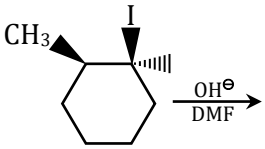
Rankers Academy JEE

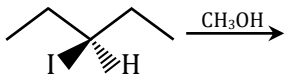
For More Join: @IITJEE_Advanced

4. Which reaction results in the formation of pair of enantiomers?

(A) 

(B) 

(C) 

(D) 

CHL161

5. Which of the following statement is true?

(A) $CH_3CH_2S^{\ominus}$ is both a stronger base and more nucleophilic than $CH_3CH_2O^{\ominus}$

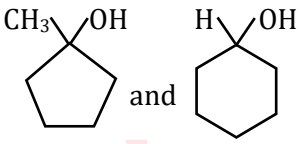
(B) $CH_3CH_2S^{\ominus}$ is a stronger base but is less nucleophilic than $CH_3CH_2O^{\ominus}$

(C) $CH_3CH_2S^{\ominus}$ is a weaker base but is more nucleophilic than $CH_3CH_2O^{\ominus}$

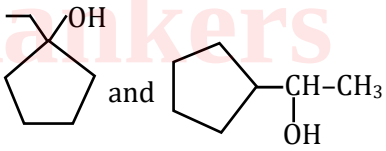
(D) $CH_3CH_2S^{\ominus}$ is both a weaker base and less nucleophilic than $CH_3CH_2O^{\ominus}$

CHL162

6. In the given pair of alcohols, in which pair second alcohol is more reactive than first towards hydrogen bromide?

(A) 

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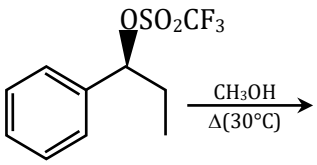
(B) 

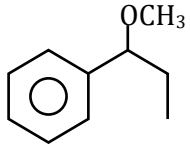
(C) $CH_3-\underset{\underset{OH}{|}}{CH}-CH_2-CH_3$ and $CH_3-CH_2-\underset{\underset{CH_3}{|}}{CH}-CH_2-OH$

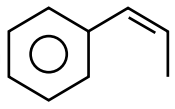
(D) $CH_3-\underset{\underset{OH}{|}}{CH}-CH_2-CH_3$ and $(CH_3)_2\underset{\underset{OH}{|}}{C}-CH_2-CH_3$

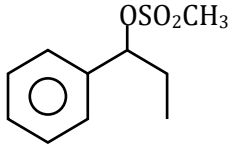
CHL163

7. Which product would be expected to predominate in the given reaction?



(A) 

(B) 

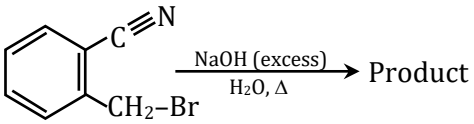
(C) 

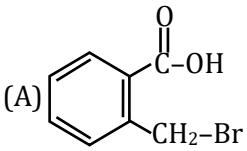
(D) None of these

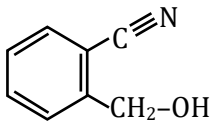
CHL164

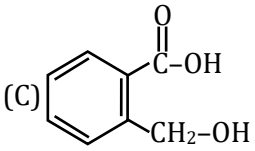
For More Join: @IITJEE_Advanced

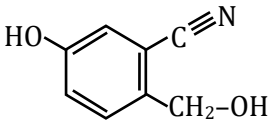
8. Which is the major product of the following reaction?



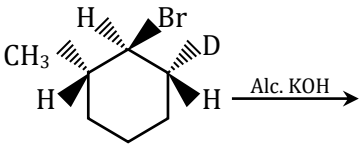
(A) 

(B) 

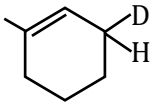
(C) 

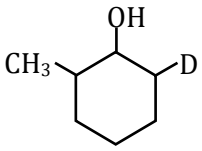
(D) 

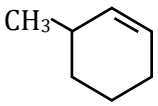
CHL165

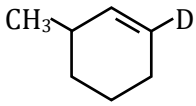
9. 

(A); Major product of this reaction is :

(A) 

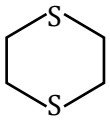
(B) 

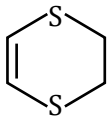
(C) 

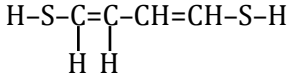
(D) 

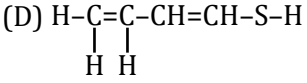
CHL199

10. 1-2-dichloro ethane + $NaSCH_2CH_2SNa \rightarrow$ (P) + 2NaCl
Unknown product (P) of the above reaction is :

(A) 

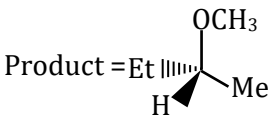
(B) 

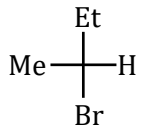
(C) 

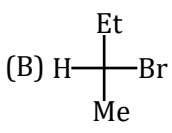
(D) 

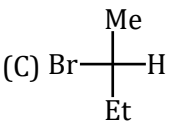
CHL167

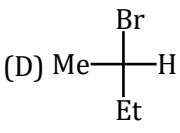
11. The back-side attack on 2-bromobutane by methoxide (CH_3O^-) gives the product shown below. Which fischer projection represents 2-bromobutane used as the reactant in this reaction?



(A) 

(B) 

(C) 

(D) 

CHL168

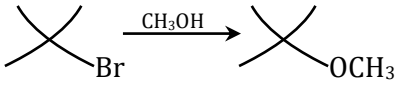
12. Which alkyl halide has maximum density?

(A) C_3H_7I (B) C_2H_5I (C) CH_3I (D) CH_3Br

CHL169

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For More Join: @IITJEE_Advanced

13. Which of the following would be true for the reaction shown?



(A) The rate of the reaction depends only on the alkyl bromide concentration.
(B) The rate of the reaction depends only on the methanol concentration.
(C) The rate of the reaction depends on both the alkyl halide concentration and the methanol concentration.
(D) The rate of the reaction depends on the concentration of neither reactant.

CHL200

Rankers Academy JEE

ANSWER KEY

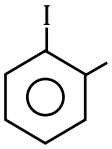
Question	1	2	3	4	5	6	7	8	9	10
Answer	A	A	C	B	C	D	A	C	C	A
Question	11	12	13							
Answer	D	A	A							

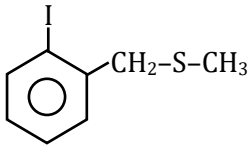
For More Join: @IITJEE_Advanced

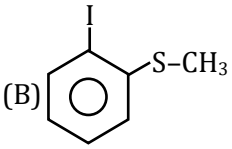
RACE
Regular Analysis Through Continuous Exercise

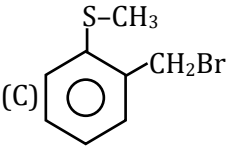
Alkyl Halides or Haloalkanes : RACE-02

Substitution Rxⁿ, Elimination Rxⁿ, SN vs E, Nucleophilicity vs Basicity, Reaction of alkyl halide

1.  $\xrightarrow{NBS}$ (A) $\xrightarrow{CH_3SNa}$ (B), Product (B) is:

(A) 

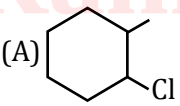
(B) 

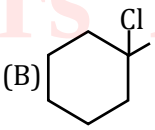
(C) 

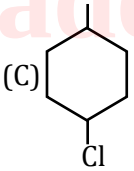
(D) None of these

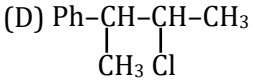
CHL172

2. S_N1 and S_N2 product are same with (excluding stereoisomer) :

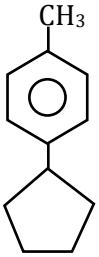
(A) 

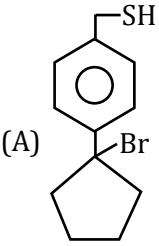
(B) 

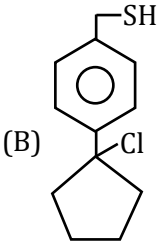
(C) 

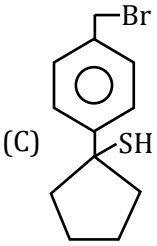
(D) 

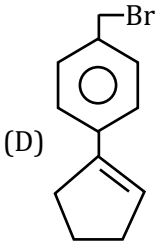
CHL173

3.  $\xrightarrow[hv]{SO_2Cl_2}$ (A) $\xrightarrow{NBS}$ (B) $\xrightarrow{KSH}$ (C), Product (C) is :

(A) 

(B) 

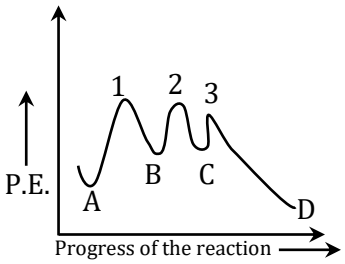
(C) 

(D) 

CHL174

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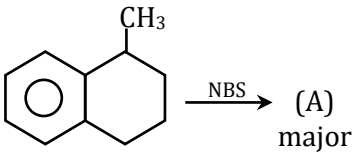
4. Energy profile diagram for an exothermic reaction, $A \xrightarrow{1} B \xrightarrow{2} C \xrightarrow{3} D$ is given below.



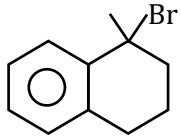
The rate determining step of the reaction is :

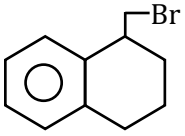
(A) $A \longrightarrow B$ (B) $B \longrightarrow C$ (C) $C \longrightarrow D$ (D) can not predict

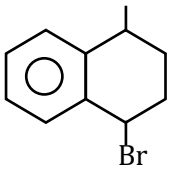
CHL175

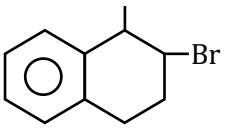
5. 

Major product is (A) is :

(A) 

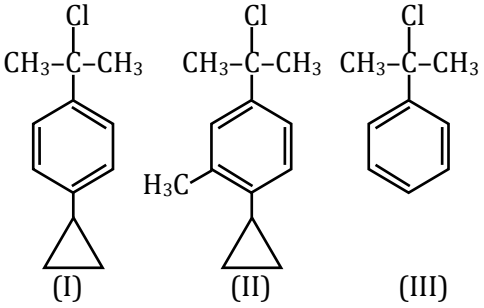
(B) 

(C) 

(D) 

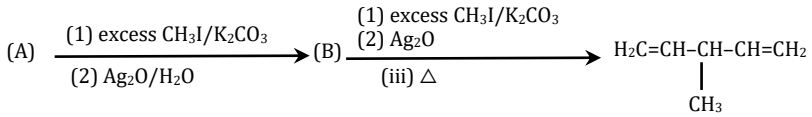
CHL176

6. The decreasing order of reactivity of the compounds given below towards solvolysis under identical conditions is :

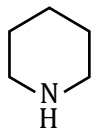


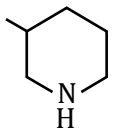
(A) II > III > I (B) I > II >III (C) III > II > I (D) II > I > III

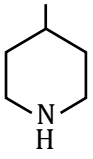
CHL177

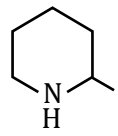
7. 

Identity A :

(A) 

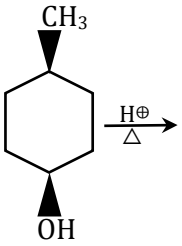
(B) 

(C) 

(D) 

CHL179

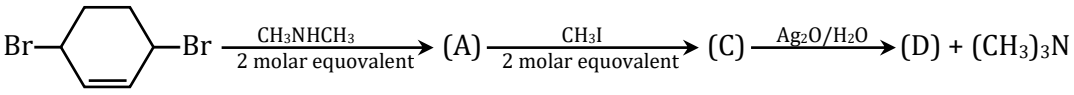
For More Join: @IITJEE_Advanced

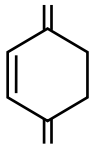
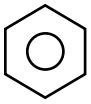
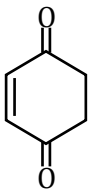
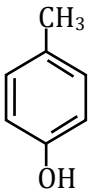
8.  ; Product obtained are :

(A) Racemic (B) Diastereomers (C) G.I (D) Positional isomers

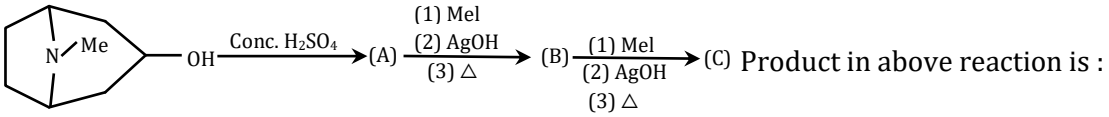
CHL180

9. End product (D) in the given sequence is :

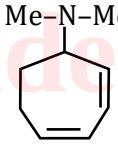


(A)  (B)  (C)  (D) 

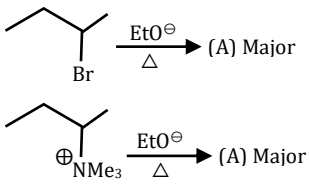
CHL181

10.  Product in above reaction is :

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(A)  (B)  (C)  (D) 

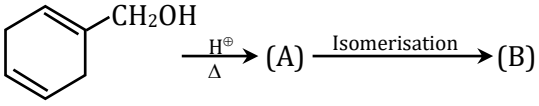
CHL182

11. 

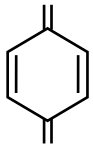
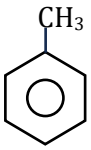
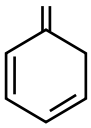
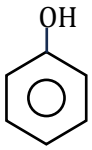
Relation between (A) and (B) is:

(A) G.I. (B) Positional isomer (C) Enantiomer (D) Chain isomer

CHL183

12. 

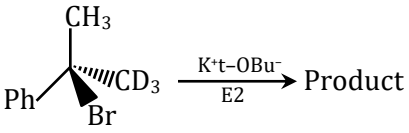
(A) on heating isomerizes to (B). What is the structure of (B) ?

(A)  (B)  (C)  (D) 

CHL184

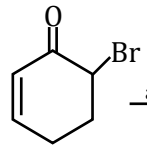
For More Join: @IITJEE_Advanced

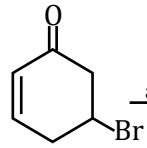
13. Which best describes the product of the following raction?

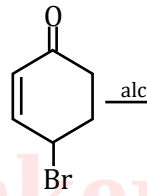


(A) Absolute configuration has been inverted
(B) Absolute configuration has been retained
(C) Racemization (loss of absolute configuration) has occurred
(D) Loss of chirality has occurred (the product is achiral)

CHL185

14. Reaction (1)  (A) (major)

Reaction (2)  (B) (major)

Reaction (3)  (C) (major)

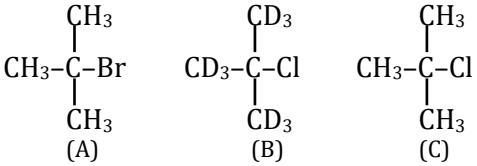
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Product obtained in above reaction (1), (2) & (3) is :

(A) A = B but C is different
(B) A =C, but B is diffrent
(C) B = C, but A is different
(D) A =B =C all product are identical

CHL186

15. Which of the following is true regatding increasing reactivity in the E2 reaction with Na^+ $C_2H_5O^-$?



(A) A is more reactive than B
(B) B is more reactive than A
(C) A is more reactive than C
(D) C is more reactive than B

CHL188

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	A	C	B	A	A	D	C	A	B	A
Question	11	12	13	14	15					
Answer	B	B	D	B	ACD					

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